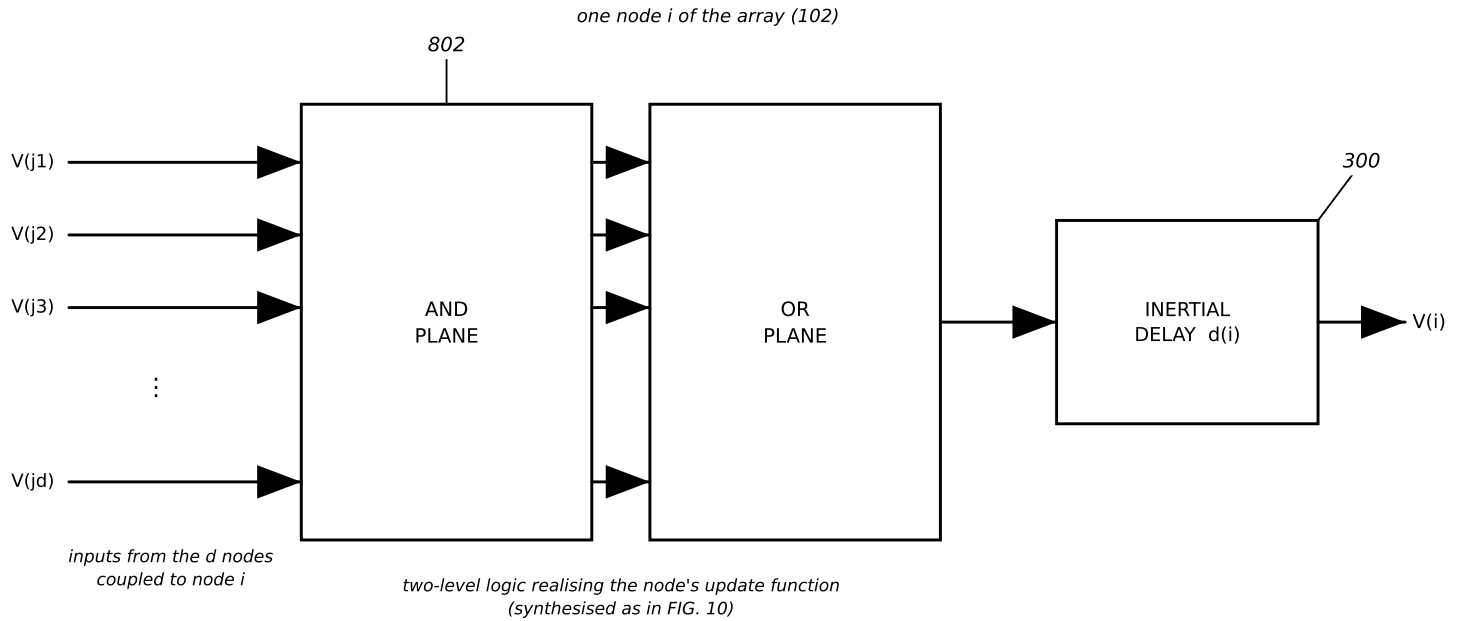


## NODE CIRCUIT AND THE DELAY SIZING RULE



$V(i)$  becomes an input to every node coupled to  $i$ , and to no other node. Node  $i$  does not receive its own output.

### SIZING RULE FOR THE NODE DELAY

$$d(i) > t_{pd, \max}(\text{node } i) \quad \text{and} \quad d(i) \neq d(j) \text{ for every coupled } j$$

*The first inequality is what makes the second one mean anything. If the node delay is shorter than the worst-case propagation delay through that node's own logic, commits interleave in an order the delay values no longer determine and the class ordering is inoperative — the array still runs, and still settles, but not under the schedule that was designed.*

*The two inequalities are independent. Satisfying only the second gives a circuit that violates its own schedule; satisfying only the first gives a correctly-timed circuit with no ordering at all.*

*The delay element is the same element as FIG. 3. When its rejection window is programmable, one element performs both functions: it sequences the node and it sets how much hazard activity is admitted.*

**FIG. 8**